CloFix

CloFix Cloud

A simple guide to every service in the console,
how FileVPS works, how it stays safe, and how you pay.

For partners, root clients and IAM users

Product: CloFix Cloud (FileVPS, storage, networking, CKS, billing)

Measured numbers: FileVPS bench and audit run, 1 November 2026

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How to read this book
Each chapter stands alone. If you are new, read chapters 1, 3, 9 and 12 first. If you are technical, add chapters 4, 6 and 7. All speed numbers are **measured** on a real test run. Nothing here is a made-up marketing figure.

Important notice for clients

Please read this page first. It explains what this book promises, what it does not promise, and what you need to do to stay safe. It helps both you and CloFix avoid surprises.

About this book

- This book is a **guide**. It is not a contract, an SLA or a guarantee. Your signed agreement, order form or plan is what applies to your service.
- CloFix Cloud keeps improving. Screens, features, limits and prices can change. Please use the latest version of this book and check your own plan.
- If this book and your agreement say different things, **your agreement is followed**.

About speed numbers

- All speed and density numbers were **measured on one lab host on 1 November 2026**.
- Your real results depend on the worker's CPU, disk, network card, your app and the load of other users. Numbers in this book are **not a promise** of the speed you will get.
- Comparison with other VMs uses typical industry ranges, not a test on the same machine.

About isolation

- FileVPS shares the worker's Linux kernel. It is **not** a KVM or QEMU hardware virtual machine.
- If your work needs its own kernel, nested virtualization, or hardware-level isolation, please tell us before you buy.
- No system is 100% secure. CloFix uses strong layers, but we cannot promise that nothing can ever go wrong.

Security is shared work

CloFix looks after	You look after
The platform, workers, controller and engine isolation	Your servers, apps, code, websites and data
Platform updates and worker kernel patches	Updates inside your own server and apps
The tools: Protection Mode, firewall, WAF, IAM	Using them: turn them on, set the rules and check them often
Keeping tenants separate by Account ID	Keeping SSH keys, PEM files, passwords and IAM keys private

Backups and your data

- VPS backups are **worker-local**. They do not include volumes, EFS or S3 mounts, and they are not off-site.
- The default S3 pool keeps **one copy**. Redis has **no** backup or restore.
- Deleting a VPS, volume or database is **permanent**. Turn on Delete Protection for important items.
- Please keep your own extra copies of important data in another place, and test your restores.
- You are responsible for the content you store and run, and for following the law.

Features still growing

- CKS is not claimed to be production-ready HA Kubernetes until live kubeadm tests are turned on for a real worker.
- Public routing of database ports is still pending. Use the Firewall and the published TLS port.
- WAF is for HTTP and HTTPS through Nginx only. It does not protect SSH, databases or other ports.

Billing

- Usage is metered every minute. Even a short run is billed for the minutes it ran, until invoiced.
- A stopped VPS or database pays for storage only.
- Please check Resource Pricing and your Due bill often. Rates shown as zero or disabled are not billed yet.

Need help?

Website: https://clofix.com | Email: support@clofix.com

1. What CloFix Cloud is

CloFix Cloud is a **cloud control panel** plus a group of Linux **workers** that run your work. You use one console to manage everything. You do not need many different tools.

Layer	What you get
Compute	FileVPS Linux servers, Library apps, Jobs and Functions, CKS (Kubernetes)
Storage	Volumes, EFS, S3 object storage, backups, managed databases
Networking	VPC, DNS, load balancers (ALB and NLB), health checks, firewall
Security	Protection Mode, SSH Login Block, Delete Protection, WAF, firewall rules, IAM, SSH keys
Platform	Dashboard, logs, invoices, wallets, resource pricing

Workers are real Linux machines. They connect to the controller with a secure mTLS link. The controller picks the best worker using live CPU, RAM and disk. It **never promises more RAM than a worker really has**.

> **Important to know**
> A FileVPS is a Linux environment that shares the worker's kernel. It is **not** a KVM or QEMU virtual machine. This is why it is so fast. Chapter 4 and chapter 11 explain what this means for you.

2. Who it is for

Role	What this person can see and do
Root client	Everything in their own account: VPS, CKS, storage, networking, IAM users and the bill.
IAM user	Only what the policy allows, for example instances, kubernetes or databases.
Partner	The client accounts they manage, and the billing for those clients.
CloFix admin	The whole platform: workers, prices, licenses and all accounts.

When you register, you get a root client account. Partners can create clients. Root and IAM users in the same account share one **Account ID**.

3. Why clients choose it

- **Speed.** A FileVPS is created in about 38 milliseconds and is usually ready in about two seconds. Normal virtual machines often need tens of seconds.
- **Density.** An idle server uses only a few megabytes of host RAM, not hundreds. All servers share one read-only Ubuntu base, so one worker can hold many of them.
- **One login for everything.** VPS, volumes, EFS, S3, backups, databases, DNS, load balancers, WAF, Kubernetes and invoices are in one console.
- **Pay for what ran.** Prices are monthly or hourly. The due bill counts every minute. A stopped VPS or database pays for storage only.
- **Safe by default.** New servers start with Protection Mode on (the root disk is read-only). CPU, RAM, swap and process limits are always enforced.
- **Separate tenants.** Every resource belongs to one Account ID. Other accounts cannot see it. One VPS cannot open a TCP connection to another VPS.

- **Kubernetes when you need it.** CKS gives you an EKS-style console. Pods run on dedicated CKS workers.

4. How much faster than other VMs

4.1 The idea in one picture

Type	How it works	What you feel
Normal cloud VM (KVM/QEMU)	Hardware, then hypervisor, then a guest kernel, then the guest OS, then your app	Boot often 15–60 s. Extra RAM cost often 100–500+ MB per VM.
CloFix FileVPS	Hardware, then the host kernel, then namespaces + cgroup v2 + OverlayFS, then your app	Create about 38 ms. Ready about 1.7 s. Idle host RAM about 1.6 MB.

FileVPS **skips the second kernel**. That is why it starts fast and uses so little RAM. It is the same kind of idea as a system container (LXC or Incus), but CloFix has its own engine, scheduler, console and billing.

4.2 Measured numbers

38 ms
Create time (median)

1.67 s
Start to ready (median)

1.6 MB
Host RAM per VPS with 100 running

3 ms
Run a command inside a guest

Measured on 1 November 2026 with the CloFix FileVPS engine and the bundled Ubuntu 24.04 root filesystem. These are real test results.

Test	Result
Create time (median)	38 ms
Start to ready (median)	1.67 s
Total launch (median)	1.71 s
Command inside the guest (median)	3 ms
Host processes per VPS	1
Extra disk after create or running	about 0.1 MB (the base OS is shared)
Apparent disk vs real disk used	about 2,335 times
100 networked VPS with SSH	PASS. 459 seconds total. About 1.60 MB host RAM each
Idle packs of 10, 25, 50, 100 VPS	About 2.56 MB host RAM each on that run
One guest reaching another on the bridge	Blocked (isolated)
25 guests running Nginx	14,213 requests per second, p95 latency 2.8 ms
CPU limit below 0.2 core	Slows down as designed
Writing past the guest disk size	Stops with an out-of-space error (ENOSPC)
Backup and restore of the writable layer	PASS
Protection Mode (`/etc` read-only)	PASS

4.3 Compared with a normal hypervisor VM

We did **not** run a same-machine test against KVM, Incus or LXC. So we do not publish a fake “CloFix vs AWS” score. The table below uses CloFix measurements and **typical industry ranges** for hardware VMs.

What you feel	Typical KVM / cloud VM	CloFix FileVPS	Gain
Time to a usable server	Often 15–60 s	About 1.7 s	About 10–35 times faster
Create call to server exists	Seconds	38 ms	Almost instant
Idle RAM per server	Often 100–500+ MB	About 1.6–2.6 MB	About 40–300 times denser
Extra disk per clone	A full guest disk	About 0.1 MB plus writes	Shared Ubuntu layer
Run a command	SSH handshake plus guest kernel	About 3 ms	Fast automation
100 small Linux servers	Heavy on CPU and RAM	100 SSH-ready guests on one worker	High density

> **Please read this honestly**
> Faster boot and higher density come from **sharing the host kernel**. A hardware VM is still the better choice if you need a different kernel, nested virtualization, or hypervisor-level isolation. The 14k requests per second result is from a lab test with 25 guests. It is not a public CDN test. Real production speed depends on your worker's cores, disks and network card. Production workers should be Linux hosts with user namespaces on (`CLOFIX_USER_NS=true`).

4.4 What fast means for a real job

Job	CloFix way	Why it is quick
New Ubuntu server	Create FileVPS	No guest kernel to boot
Scale a website	ALB plus restore extra FileVPS from a backup	Overlay clone, not a full disk copy
Run a script	Jobs and Functions	Fresh overlay, no VPS to set up
Deploy WordPress or Node	Library Quick App	A sandbox, not a VM install
Kubernetes workload	CKS on dedicated workers	kubeadm nodes with an EKS-style console

5. Console map

Use the search box in the console. This table shows where things live.

Group	Pages
Overview	Dashboard: VPS and worker counts, CPU, RAM, disk, swap, and network charts
Compute	FileVPS, Library, Jobs and Functions, CKS
Storage	Volumes, EFS, Object Storage (S3), Backups, Databases
Networking	Load Balancers, VPC, DNS Service, Health Checks, Firewall
Security	Security (Protection Mode, SSH, WAF, ports), Firewall, IAM, Users (admin and partner)
Platform	Log Service, Workers (admin), Billing and Payments, Settings

IAM permissions follow these same groups. Root and IAM users always stay inside one client account.

6. Compute

6.1 FileVPS

A FileVPS is a Linux server that runs on a CloFix worker.

- **Image:** bundled Ubuntu 24.04. Servers using the same image share a read-only base. Each server has its own writable OverlayFS layer.
- **Limits:** CPU, RAM, swap and process counts use cgroup v2. These are limits on shared host resources, not dedicated physical cores.
- **Placement:** the scheduler uses live free CPU, RAM and disk. It does not overcommit RAM.
- **Login:** SSH key (PEM), password (PAM), or both. You can generate a PEM when you create the server. Download it **once**.
- **Network:** private address on the worker bridge `10.231.0.0/16`. In local lab setups: SSH `22000–22099`, HTTP `23000–23099`, HTTPS `23100–23199`, extra maps `24000–24099`.
- **Details page:** CPU, RAM, swap, network in and out, port mappings, services, WAF, volumes and S3 mounts.
- **Instances list:** shows FileVPS and CKS nodes together (see the Kind column). A CKS row opens the cluster. It is not a FileVPS guest.

> **Protection Mode (on by default)**
> The guest root disk is read-only. `/run`, `/tmp`, `/var/log` and the APT cache are temporary and clear on restart. `apt update` still works. To install packages you must turn Protection Mode off. Turning it on or off restarts a running VPS. It does not stop your processes or network. It is **not** a replacement for a hardware VM.

6.2 Library (Quick Apps and VPS software)

Quick Apps run in a managed sandbox. You do not need your own VPS or SSH box.

- Adminer, Uptime Kuma, Jenkins
- ZIP websites: static HTML, PHP, Node.js, Next.js, WordPress
- You can attach an app to an existing managed database

- From the Library you can open, log in, start, stop, view logs, export data and delete

Apps use namespaces and cgroup v2, run as user 65534, use a read-only template and a private network, and keep data only in `/data`.

VPS software installs packages into a running FileVPS (Protection Mode must be off). This includes Java services such as WildFly or Tomcat where they are in the catalog.

6.3 Jobs and Functions

Run Python, Go, Node, PHP, Java, Ruby, Bun or Deno **without making a VPS**. Each run gets a fresh overlay, an unprivileged user, resource limits and a private network. Each worker shows which runtimes it has.

6.4 CKS – CloFix Kubernetes Service

CKS is the Kubernetes product in the console (`/kubernetes`, also `/cks`). Think of it as **EKS-style operations on CloFix workers**.

Area	What you can do
Clusters	Create, rename, delete protection, download kubeconfig
VPS groups (node pools)	Sized like FileVPS (CPU, RAM, disk) on dedicated CKS workers
Workloads	Namespaces, pods, Deployments, Services, Ingress, Secrets, ConfigMaps, PVC, Jobs, HPA, NetworkPolicy
Helm	Install, upgrade, uninstall
CloFix attach	Bind an existing load balancer, FileVPS, volume, EFS or S3 into the cluster
Add-ons	Local StorageClass, nginx IngressClass, RBAC YAML
Operations	etcd backup and restore, rolling upgrade, HA with 3 or 5 control planes
DNS	Automatic `*.cks.` names for the cluster, Services and Ingresses
Billing	CKS control-plane unit plus node CPU, RAM and disk. Attached load balancers stay on the LB row

Hard limits – read before you buy Kubernetes
FileVPS guests **cannot** run kubelet (shared kernel, seccomp, no cgroup delegation). CKS nodes are **dedicated worker machines**, not FileVPS. The Kubernetes API stays on the worker. CloFix does **not** publish port `6443` on the internet. `production_ready` stays **false** until live kubeadm HA is turned on for a real kubeadm host. The local Compose setup uses a lab executor for demos. That is not production Kubernetes. There is no pod exec or terminal, no public kube-apiserver, and the network plugin is Flannel (not Cilium).

7. Storage

Pick the right product first. They look alike in the menu, but they are not the same.

Use this	When	Where it lives	Attach to
Volume	One VPS needs a long-lasting disk (`/mnt/data`, database files)	One worker directory, or a capped ext4 disk	Same worker only, one to one
EFS	Many VPS share one web folder, even on other workers	CloFix S3 prefix, mounted on each worker	Any worker, many VPS
S3	Object files, or mount a bucket as a folder	CloFix worker pool, or AWS / GCS / Azure gateway / custom	Mount with s3fs into a VPS
VPS backup	Clone or recover a guest's writable layer	Compressed file on the same worker	Restore as a new VPS on the same worker
Database service	Managed MySQL, PostgreSQL, MongoDB or Redis	Isolated project on a database-capable worker	Connect over TLS, optional DNS name

Volume, EFS and S3 mounts **must not use the same path** in a guest. A running VPS **restarts** when you attach or remount storage. Even with Protection Mode on, a mounted volume, EFS or S3 path stays writable.

7.1 Volumes (/volumes)

Long-lasting worker storage that is separate from the VPS overlay layer.

Action	What happens
Create	Give a name and pick a worker. An optional **hard size limit (MB)** makes a capped ext4 image. Leave it blank for unlimited use of that worker's disk.
Attach	Only to a VPS on **that** worker. Default path `/mnt/data`. A running VPS restarts.
Detach	Data stays. You can attach it again later.
Delete	Removes the data from the worker for good.
Edit	Display name only. The size limit is fixed when you create it.
Explorer	Browse, upload files and folders (4 MB per file), make folders and text files, download folders, move. Text edit up to 2 MB.

Good: survives VPS delete if you detach first; stays writable under Protection Mode; a size limit stops one guest from filling the worker. **Limits:** not shared across workers; not included in a VPS backup; delete is permanent.

Tip: if you mount an *empty* volume at `/var/www/html`, it hides the image's web files. Nginx may show 403 until you upload a readable `index.html` in Volume Explorer.

7.2 EFS (/efs)

One **named share** that many VPS can mount, including servers on other workers. Files live on CloFix S3. Each worker mounts the prefix once and shares it with attached guests.

Typical use: a load-balanced website. Attach the same share to every member at `/var/www/html`. Creating a load balancer does **not** create EFS for you. Pick an existing share on the LB form.

Action	What happens
Create share	Name only. Uses CloFix S3 by default. Works on any worker.
Attach VPS	Any worker. Default `/var/www/html`. Must not overlap a volume or S3 mount. A running VPS restarts.
Rename	Changes the display name. Attachments and the S3 prefix stay the same.
Delete	Detach every VPS first. The S3 prefix and objects are **not** deleted.
Explorer	Upload once, every attached VPS sees it. 4 MB per file. Folders, text files, drag and drop, download folder, move, edit text up to 4 MB.

Good: upload once and every replica sees it; works across workers. **Limits:** it is not a full NFS appliance; it is backed by S3; the Explorer has a file size limit.

7.3 Object Storage / S3 (/storage/s3)

- **CloFix S3 (default shared pool).** Each Linux worker adds `/var/lib/clofix-filevps/s3-volume`. The page shows online storage workers, total disk and free space (SeaweedFS keeps about 5% free).
- **External S3.** Amazon S3, GCS S3 interoperability, an Azure S3-compatible gateway, or any custom endpoint. Credentials are **encrypted at rest and never shown again** in the browser.

Action	What happens
Bucket files	Browse, search, upload, download folder, make folders and text files, move, delete. Text edit up to 2 MB.
Mount in a VPS	Uses s3fs on the worker (`/dev/fuse`). Default guest path `/var/www/html`. A running VPS restarts. System paths are blocked.
View mounts	Unmount before you delete a connection.

> **Honest limits**
> Pool placement `000` means **one copy only**. If a worker disk fails, objects on that worker can be lost. Capacity is the size of the backing disk, not a reserved quota. The local Compose S3 gateway only listens on loopback. Production needs TLS and a private network (do not publish cluster ports 9333/19333 and 8080/18080). Docker Desktop on macOS is not a production storage node.

7.4 VPS backups (/backups)

A worker-local snapshot of a **stopped** FileVPS writable root. To restore, choose **Run backup as a new VPS**. The new VPS gets its own SSH key, CPU, RAM, disk and swap.

Included	Not included
Compressed writable-layer archive	Persistent **volumes**
Packages and files written on the overlay	**S3** and **EFS** mounts

Included	Not included
Restore on the same worker using the cached base image	Restore on a different worker
Image source for load balancer scale-out	Snapshot of a running VPS

The size shown in the UI is the **compressed archive**. The disk set on restore is for placement counting, not a guest disk quota. Make backups only while the VPS is stopped.

Good: quick clone of a ready app image; LB members restore from the same backup; a new SSH key on restore. **Limits:** not a crash-consistent multi-disk backup; not off-site unless you copy the archive yourself; the original worker is required.

7.5 Database service (/databases)

Managed database **instances**. This is not a MySQL package inside a FileVPS.

Engine	What to know
MySQL	Logical databases, users and grants. TLS required. Backup uses `mysqldump --single-transaction` (InnoDB). Triggers, routines and events are not fully backed up.
PostgreSQL	TLS only (unencrypted connections are rejected). Custom-format `pg_dump`.
MongoDB	TLS on the published port. Native compressed archives. `fsyncLock` is used during backup.
Redis	TLS on the published port. Logical databases are **key prefixes** in a shared instance. **Backup and restore are not available.**

In the console you can manage instances, logical databases, users, **encrypted** backups (central and S3, resumable upload), schedules and live jobs. The create form asks for name, engine, worker, VPC, version, CPU, RAM and disk. You can download the instance **TLS CA**. A failed setup can be retried.

Runtime (on workers where the operator turned it on): each project gets its own container, private network, CPU/RAM/process limits, a pre-made ext4 disk, encrypted admin credentials, ports `26000–26099`, and up to **128** connections. You cannot pick random images. The controller never gets a Docker socket.

Backup and restore: streams use AES-256-GCM. To restore into an existing destination you need a fresh **safety backup** (under 10 minutes old, matching checksum). A full replace or table cutover can be destructive and is **not** an automatic rollback. If it fails, recover from the safety backup. Engine versions must match.

Good: isolated engine, TLS, IAM, encrypted backups, schedules, no SSH into a database server. **Limits:** needs a database-capable worker; no Redis backup; public routing of database ports is still pending (use Firewall plus the published TLS port); a stopped instance pays **storage only**.

8. Networking

VPC. Private IPv4 on the FileVPS bridge `10.231.0.0/16`. The default block cannot be deleted. Extra blocks must not overlap. New VPS, apps and databases use the default VPC unless you choose another.

8.1 DNS Service (/dns)

The controller serves a DNS zone, by default `clofix.internal`. This is CloFix DNS, not a public worldwide network. You can download a **BIND-style zone file** and point your real nameservers at it. An optional **publish address** fills load balancer records that have no IP yet.

Automatic records (switch them in Zone settings): FileVPS, load balancers, CKS clusters, databases, managed apps and workers.

Source	Example name
FileVPS	`{id}.vps.clofix.internal`
Load balancer	`{id}.lb.` or `{id}.alb.clofix.internal`
Database	`{id}.db.clofix.internal`
Managed app	`{id}.app.clofix.internal`
Worker	`{id}.worker.clofix.internal`
CKS cluster	`test-cluster.cks.clofix.internal`
CKS Service	`nginx.svc.test-cluster.cks.clofix.internal`
CKS Ingress	`nginx.ing.test-cluster.cks.clofix.internal`

Manual records can point to a FileVPS, LB, CKS cluster, Service or Ingress, database, app, worker, or a manual address. Types: A, AAAA, CNAME, TXT. A **port** is stored and exported as an **SRV** record. The Port list shows mapped HTTP, HTTPS, lab, NodePort and guest ports, not only 80.

Good: one zone for VPS, LB, CKS and databases; export to your production DNS. **Limits:** it does not replace Route 53 or Cloudflare worldwide unless you publish the zone. Port 53 in Compose is local only.

8.2 Load balancers

- **ALB** – HTTP and HTTPS. Use the path `/lb//` or a custom domain that points to the controller.
- **NLB** – TCP. Gets a unique IP from the Settings → NLB pool.
- The LB owns backend FileVPS servers restored from a **ready VPS backup**. It scales by health, CPU and (optionally) hits.
- Optional **EFS share** is mounted on every member (default `/var/www/html`).
- Algorithms: round robin, least connections, least response time, IP hash, URI hash, random, power of two.

8.3 Health checks and firewall

Health checks: HTTP, TCP and ICMP, plus worker and LB availability. Alerts go to the channels in Settings. This is CloFix's own checker. Uptime Kuma is a separate Library app.

Firewall: inbound rules for FileVPS port forwards, managed-app gateways and database TLS ports. See chapter 9.

9. Security and VPS security

Think of CloFix security as **two layers**:

- **Engine isolation** – what the FileVPS engine always does (namespaces, seccomp, cgroup, isolated bridge).
- **VPS controls** – what you switch on for each server in **Security**, **VPS Details** and **Firewall**.

Open **Security** (`/security`) for a fleet view: how many VPS have Protection Mode on or off, SSH Login Block, Delete Protection, WAF installed, custom port forwards, and active workers.

9.1 FileVPS isolation (always on)

FileVPS uses **OS-level isolation**, not a hypervisor. Guests share the worker kernel. Isolation comes from Linux namespaces, cgroup v2, OverlayFS, veth and nftables, and seccomp.

Layer	What the guest gets
Namespaces (mount, PID, UTS, IPC, net, cgroup)	Its own file view, process list, hostname, IPC, network and a hidden cgroup tree
User namespace (production default `CLOFIX_USER_NS=true`)	Guest root maps to a unique host user range (default base `100000`), not host root
Seccomp and `no_new_privs`	Blocks `mount`, `umount2`, `unshare`, `pivot_root`, `bpf` and similar calls
Dropped capabilities	No host-admin powers. `CAP_SYS_ADMIN` and `CAP_NET_ADMIN` stay dropped
Devices	No `/dev/mem`, `/dev/kmem`, `/dev/port` or block devices. Making block devices is denied
OverlayFS	Shared read-only Ubuntu base, own writable layer per VPS
Network	Private `10.231.0.0/16` veth. Each veth is bridge-isolated. nftables forward is default-drop
Resources	CPU, RAM, swap and `pids.max` per VPS. One guest cannot starve the host without limit

Measured in the 1 November 2026 audit: files and processes of other VPS are invisible; TCP between VPS is blocked; sensitive `/proc` and `/sys` paths are not writable; `dmesg` cannot be read; host runtime folders are hidden; with Protection Mode `/etc` is read-only (EROFS). Local Compose may use `CLOFIX_USER_NS=false` for Docker Desktop. That is a lab setup and weaker than production.

> **Not claimed**
> There is no per-guest AppArmor or SELinux profile. A bug in the host **kernel** can affect every FileVPS on that worker. The flip side: patch the worker kernel and every guest is patched too. This is not KVM hardware isolation.

9.2 Controls you set on each VPS

These are on **VPS Details → Security** and **Security → VPS controls**. Changing an SSH policy restarts a running VPS.

Control	Default	What it does	Tip
Protection Mode	On for new VPS	Guest root is read-only. Temporary folders clear on restart. Installing packages needs it off.	Keep on for production sites. Turn off only to install, then turn on again.
SSH Login Block	Off	Blocks **new** SSH logins. Existing sessions may stay until they end.	Use after deploy or during an incident.
Delete Protection	Off	Blocks delete through CloFix APIs and worker actions. A host admin with disk access can still remove files.	Turn on for important servers and CKS clusters.
SSH keys / password	Your choice	Paste a public key, or generate a PEM (shown once). Password login uses PAM.	Prefer keys. Never reuse the one-time PEM after download.
Custom port forwards	SSH, HTTP, HTTPS mapped	Extra guest ports on unique host ports. Each forward **exposes** that service.	Review Security → Port forwards. Remove unused ones.
Inbound security group	Open until you restrict	CIDR allow or deny for ports already mapped. Existing connections stay allowed.	Set it in Firewall → VPS.

9.3 OWASP CRS WAF (web firewall)

Install it from VPS Details or from the Library (catalog ID `owasp-waf`). It is Ubuntu Nginx plus ModSecurity plus the OWASP Core Rule Set, in **blocking mode**. Install, uninstall and rule changes need Protection Mode **off**. Nginx config is checked with `nginx -t` and then reloaded. Uninstall removes ModSecurity and CRS but **leaves Nginx**.

Rule group	Covers
HTTP protocol attacks / enforcement	Protocol and RFC checks (CRS 920xxx)
Allowed HTTP methods	CRS 911xxx
Scanner detection	CRS 913xxx
IP reputation	CRS 910xxx (state resets when the VPS restarts)
Layer 7 DoS	CRS 912xxx
Multipart / uploads	CRS 922xxx
LFI / RFI	Local and remote file inclusion
RCE	OS and command injection
PHP injection	PHP attack rules
Generic application attacks	Generic CRS attack tags
XSS	Cross-site scripting
SQLi	SQL injection
Session fixation	Session fixation
Response data leakage	Disclosure rules

The **WAF error log** in the console shows recent Nginx and ModSecurity blocks from the worker heartbeat.

> **Scope – very important**
> The WAF checks **HTTP and HTTPS through this VPS's Nginx only**. It does **not** protect SSH, database ports, other guest ports, your app's login logic, or traffic that skips Nginx (for example a custom port forward straight to the app). CRS can block real users, so test your app after you turn on groups. Protection Mode and WAF are **separate layers**.

9.4 Firewall and published ports

The **Firewall** (`/firewall`) filters packets for CloFix published ports. It is not the HTTP WAF.

Scope	What it filters	What it does not
Workers	FileVPS port forwards, managed-app gateways, database TLS ports	Host SSH 22 and admin ports
VPS	Already-mapped guest ports (security group)	Ports you have not published. You must still map a port to expose it

Rules can allow or deny, for TCP, UDP, ICMP or any, with an optional port or range and an **IPv4 CIDR**. Shortcuts: `0.0.0.0/0` or the same VPC. By default everything is allowed so localhost and Compose keep working, until you tick **Restrict inbound**.

> **Good setup for a web VPS**
> Map only ports 80 and 443 (CloFix already does this). Restrict SSH to your office or VPN CIDR. Leave Protection Mode on. Install the WAF on Nginx.

9.5 IAM, tenancy and platform

Control	What it does
Account ID	Root and IAM users of one client share one tenant. Other accounts cannot see the object.
IAM	Users, groups, policies, access keys and bearer tokens. `security:manage` covers Protection Mode, SSH block, delete protection, WAF rules and port forwarding. It is separate from `instances`, `kubernetes`, `billing` and others.
Cookie session	Console login. IAM keys are used for the API.
Worker mTLS	One-time enrollment token plus controller CA. No enrollment secret sits in Compose files.
Encrypted secrets	Kubeconfig, CKS jobs, database admin passwords and backup snapshots
Library installs	Fixed service IDs (Nginx, WAF, WordPress and so on), not free shell commands
Jobs and Functions	Unprivileged user, fresh overlay, private network, no extra VPS
Managed apps	User 65534, read-only template, seccomp, private network, persistent `/data` only
CKS API	The Kubernetes API stays on the worker. CloFix does not publish `6443`

9.6 Recommended VPS security steps

1. Create the FileVPS with an **SSH public key** and Protection Mode **on**.
2. Download the PEM once if you generated one. Store it offline.
3. Install packages with Protection Mode **off**, then **turn it back on**.
4. Install the **OWASP CRS WAF** only if this VPS serves HTTP through Nginx. Turn on SQLi, XSS, RCE and LFI first. Add stricter groups after testing.
5. In **Firewall → VPS**, limit SSH to known CIDRs. Do not leave extra port forwards.
6. Turn on **Delete Protection** for production servers.
7. Use **SSH Login Block** when nobody needs to log in (app-only servers).
8. Use Library Quick Apps or managed databases when you do not need a full guest OS.
9. Check the **Security** overview every week: Protection Mode off, WAF missing, extra forwards.

9.7 Honest limits

- Shared kernel: not a hardware VM and not nested KVM.
- No per-guest AppArmor or SELinux profile in the current engine.
- WAF, firewall and Protection Mode are three different things. Use all three for a public website.
- Host administrators with disk access can still read worker storage. Delete Protection is an API and UI lock.
- Compose with `user_ns=false` is weaker than production with `user_ns=true`.
- The isolation audit ran on FileVPS on linuxkit. It is not a kernel exploit assessment.

10. Billing and plans

Resource Pricing has Standard and Low-cost rates, monthly and hourly, for CPU, RAM, SSD, CKS, load balancer, backup, volume, EFS, S3 and applications. CKS is the control-plane unit. Node CPU, RAM and disk use the CPU, RAM and SSD rows. Attached load balancers bill on the Load balancer row. A rate of zero (or disabled) means that unit is not billed yet. Admins manage catalog plans, unit rates, discounts, gateway and currency or tax in **Settings → Pricing and Plans**.

Billing and Payments shows invoices, wallets, manual payments, subscriptions, refunds and credits.

- **Live resources:** VPS, CKS clusters, LBs, apps, volumes, backups, S3 and EFS, with a USD and BDT estimate.
- **Due bill:** the hourly meter runs every minute. A 5 or 10 minute run is billed for those minutes even if you delete the resource afterwards, until it is invoiced.
- **Stopped VPS or database:** storage only.
- **Invoice PDF:** CloFix brand, bill-to, period, lines, tax, in USD and BDT.
- **Cards:** card charges are **not** processed in this stack. Gateway secrets are stored and never returned by the API.

License quotas can limit VPS, CKS clusters, nodes and pools, load balancers, DNS and backups. If a CKS limit is not set, it stays unlimited so older plans keep working.

11. What CloFix is not

Please tell technical buyers this at the start.

- FileVPS is **not** a KVM/QEMU virtual machine and **not** nested hardware virtualization.
- FileVPS is **not** a Kubernetes node. CKS needs dedicated workers with kubeadm and containerd.
- Local Docker Compose is a **lab**. The fake CKS on Compose is for UI and API practice, not production HA Kubernetes.
- Backups are **worker-local** writable layers, not enterprise multi-disk crash-consistent backups.
- S3 default is **one replica**.
- CloFix does not issue public TLS certificates for ALB custom domains.
- DNS in the controller is a **CloFix zone**, not a public worldwide DNS network, unless you export the zone to your nameservers.
- Density and requests-per-second numbers come from **one lab host**. Your production worker's cores, disks and network card set the real limit.
- FileVPS security is namespaces, seccomp, cgroup, Protection Mode, WAF and firewall – **not** a hardware hypervisor.

12. Getting started

1. Sign in to the CloFix Cloud console (admin bootstrap, or **Register** for a root client).
2. Open **FileVPS → Create FileVPS**. Keep Protection Mode on unless you must install packages.
3. Save the PEM or password when it is shown. The private key appears only once.
4. Open **Security** and **Firewall**. Check Protection Mode, optional WAF, SSH CIDR and unused port forwards.
5. Open **Instances**. FileVPS and CKS nodes appear together. Search `cks` or `eks` to filter Kubernetes nodes.
6. Add **storage** as needed: Volume (one VPS, same worker), EFS (shared web folder or LB), S3 (objects plus optional s3fs mount).
7. Stop a golden FileVPS and make a **backup** before you scale an ALB from that image.
8. Create a **database instance** (MySQL, PostgreSQL, MongoDB or Redis) on a database-capable worker. Download the TLS CA.
9. Open **DNS**. Check the automatic records for VPS, LB, CKS and database. Add a friendly hostname if you want.
10. Put HTTP in front of FileVPS with a **load balancer** from a ready backup. Attach EFS if members share a web folder.
11. For Kubernetes, open **CKS**, create a cluster on a dedicated worker, then **Add VPS group** for the nodes that run pods.
12. Watch **Billing → Due bill** and **Live resources** so the rates match what you turned on.

Operators enroll Linux workers with a one-time token and the controller CA. Clients never handle worker enrollment.

13. Glossary

Term	Meaning
FileVPS	CloFix Linux guest with OS-level isolation (namespaces, cgroup v2, OverlayFS)
Worker	A Linux host enrolled to the controller that runs FileVPS, apps, databases and CKS
CKS	CloFix Kubernetes Service – EKS-style clusters on dedicated workers
VPS group	A CKS node pool. Nodes are sized like FileVPS but are not FileVPS guests
Protection Mode	Read-only guest root filesystem (on by default)
SSH Login Block	Blocks new SSH logins to that VPS
OWASP CRS WAF	ModSecurity Core Rule Set on the guest's Nginx HTTP and HTTPS
Volume	Worker-local disk, attached one to one to a FileVPS on that worker
EFS	Shared S3-backed folder for many VPS, workers and LB members
S3	Object storage (CloFix pool or external) with an optional s3fs mount
VPS backup	Snapshot of a stopped writable layer. Restore as a new VPS on the same worker
Database service	Managed MySQL, PostgreSQL, MongoDB and Redis with TLS
DNS Service	Controller zone (`clofix.internal`) with automatic and manual records and BIND export
VPS security group	CIDR allow or deny for ports that are already mapped
ALB / NLB	HTTP load balancer / TCP load balancer
Due bill	Minute-level hourly usage waiting to be invoiced
Account ID	Tenant key shared by Root and IAM users of one client

CloFix Cloud Client Book. Performance figures come from the FileVPS bench and audit run of 1 November 2026. VM boot and RAM ranges for other clouds are typical industry figures. CKS brings an EKS-style Kubernetes console to dedicated CloFix workers, so your teams can run clusters with the tools they already know.